



Question Set

Question 1

Consider an EURUSD forward and option book collateralized in USD. Define the state variables (spot FX, domestic discount factors, foreign discount factors, and volatility) and derive the covered-interest-parity forward and domestic-currency option value. State the measure, filtration, and tradability assumptions.

$$F_{d/f}(0, T) = S_0 \frac{P_f(0, T)}{P_d(0, T)}$$

Question 2

For an EURUSD forward and option book collateralized in USD, explain which probability measure is natural for pricing. Write the density-process logic needed if the starting model is under the physical measure.

Question 3

Identify the discounted tradable or density-adjusted process in an EURUSD forward and option book collateralized in USD. Prove the martingale condition and name one modelling choice that would break it.

Question 4

Using spot 1.0850, USD rate 4.2 percent, EUR rate 2.8 percent, and maturities 3M to 2Y, construct a two-step numerical calculation. Show intermediate state values before discounting or aggregating.

$$F_{d/f}(0, T) = S_0 \frac{P_f(0, T)}{P_d(0, T)}$$

Question 5

For an EURUSD forward and option book collateralized in USD, compute and interpret spot delta, forward points sensitivity, and foreign-rate exposure. Explain the hedge sign and the unit of each sensitivity.

Question 6

Suppose the desk observes ATM vol, risk reversal, butterfly, and forward points. Write the calibration objective, the weights you would use, and the parameter-identifiability checks.

Question 7

State the terminal and boundary conditions for an EURUSD forward and option book collateralized in USD. Explain how domestic payoff conversion and zero/large spot limits prevents economically impossible prices.

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Question 8

Design a Monte Carlo or lattice estimator for an EURUSD forward and option book collateralized in USD. Specify risk-neutral FX paths under the domestic measure, the estimator, the discounting rule, and the standard error.

Question 9

A junior quant implements an EURUSD forward and option book collateralized in USD and gets a stable but wrong answer. Explain how using the wrong discount curve or reversing the currency quotation creates the error and design a diagnostic test.

Question 10

Write an NumPy algorithm for an EURUSD forward and option book collateralized in USD. Include inputs, outputs, diagnostics, and one benchmark case.

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Question 11

Starting from the payoff definition of an EURUSD forward and option book collateralized in USD, derive a replication, dynamic-programming, or conditional-expectation representation without skipping the economic argument.

Question 12

Create a stressed-market version of spot 1.0850, USD rate 4.2 percent, EUR rate 2.8 percent, and maturities 3M to 2Y. Recompute the value or risk metric and explain which term dominates the change.

Question 13

Use a second-order Taylor approximation for an EURUSD forward and option book collateralized in USD. Separate linear exposure, convexity exposure, cross terms if relevant, and residual model risk.

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Question 14

Show how a change of numeraire or change of measure would simplify one expectation in an EURUSD forward and option book collateralized in USD. Identify what becomes a martingale after the change.

Question 15

Give a proof-style answer showing when the wealth process for an EURUSD forward and option book collateralized in USD is self-financing. Explain why predictable trading strategies matter.

Question 16

Market quotes for ATM vol, risk reversal, butterfly, and forward points are noisy. Propose a robust calibration workflow with initial guesses, constraints, residual plots, and out-of-sample checks.

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Question 17

For an EURUSD forward and option book collateralized in USD, compare the mathematical boundary condition with the trader's intuition. Give one limiting case where the answer should be obvious.

Question 18

Design a convergence study for risk-neutral FX paths under the domestic measure. Specify path counts, time-step refinement, confidence intervals, and the exact quantity plotted.

Question 19

Find one way that using the wrong discount curve or reversing the currency quotation could pass a superficial unit test. Write a stronger test that would fail for the biased implementation.

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Question 20

Explain an EURUSD forward and option book collateralized in USD and code the core routine. Give the derivation outline, pseudocode, and three numerical sanity checks.